AGENT DESIGN ASSESSMENT

CO1_AT1

MLA 0104 AI and Expert System

Submitted by

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Under the Supervision of

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ABSTRACT

Artificial Intelligence (AI) has emerged as one of the most influential technologies in the field of computer science, enabling machines to perform tasks that traditionally require human intelligence. In recent years, AI has significantly transformed various sectors, including healthcare, banking, transportation, manufacturing, and education. One of the most effective applications of AI in education is the development of intelligent conversational agents, commonly known as chatbots, which provide instant support and improve communication between students and educational institutions.

The AI-Based Student Support Assistant is an intelligent chatbot developed using **Google** Dialogflow, a cloud-based conversational AI platform. The primary objective of this project is to provide immediate and accurate responses to students' academic and campus-related queries. The chatbot assists students by answering frequently asked questions regarding course registration, class timetables, examination schedules, attendance details, library services, hostel facilities, transportation, and other campus support services.

The chatbot **utilizes** Natural Language Processing (NLP)**,** Natural Language Understanding (NLU)**,** Intent Recognition**,** Entity Extraction**, and** Context Management to understand user queries expressed in natural language. When a student submits a question, the Dialogflow engine analyzes the input, identifies the user's intent, extracts relevant information, and generates an appropriate response based on predefined intents and entities. The chatbot is designed to maintain conversational context, allowing it to provide meaningful follow-up responses during multi-turn interactions. If a user's query does not match any predefined intent, the Default Fallback Intent politely requests the user to rephrase the question, ensuring a smooth conversational experience.

In conclusion, the AI-Based Student Support Assistant Using Dialogflow demonstrates the practical application of Artificial Intelligence in modern education. The project showcases how conversational AI can simplify student support services by automating repetitive tasks and providing intelligent assistance in real time. Future enhancements may include multilingual support, voice-based interaction, real-time database integration, personalized responses based on student profiles, and integration with college ERP systems. These improvements will further enhance the chatbot's capabilities and contribute to the development of smarter, more efficient, and student-friendly educational environments.

INTRODUCTION

Artificial Intelligence (AI) is one of the most rapidly growing technologies that enables computers to perform tasks requiring human intelligence, such as learning, decision-making, and understanding natural language. AI is widely used in healthcare, banking, transportation, business, and education to improve efficiency and automate routine tasks.

In the education sector, students frequently require information about course registration, class timetables, examination schedules, attendance, and campus services. Traditionally, these queries are handled manually by administrative staff or faculty members, which can be time-consuming and lead to delays, especially during busy academic periods.

To overcome these challenges, educational institutions are adopting AI-powered chatbots that provide instant and accurate responses to student queries. A chatbot is a conversational software application that uses Natural Language Processing (NLP**)** and Natural Language Understanding (NLU**)** to understand user input and generate appropriate responses. These chatbots are available 24×7 and help reduce the workload of administrative staff while improving the overall student experience.

This project presents an AI-Based Student Support Assistant developed using Google Dialogflow**.** Dialogflow is a cloud-based conversational AI platform that enables developers to create intelligent chatbots using intents**,** entities**,** training phrases**, contexts**, and responses**.** It analyzes user queries, identifies the user's intent, and provides relevant information based on predefined knowledge.

The chatbot developed in this project assists students with information related **to** course registration, class timetables, examination schedules, attendance details, library services, hostel facilities, and other campus services**.** It recognizes different ways of asking the same question and responds accurately. If a query is not recognized, the chatbot uses the Default Fallback Intent to politely ask the user to rephrase the question.

The AI-Based Student Support Assistant demonstrates how Artificial Intelligence can improve communication between students and educational institutions by providing quick, accurate, and user-friendly support. The system reduces response time, minimizes manual effort, and offers a scalable solution that can be further enhanced with multilingual support, voice interaction, and real-time database integration.

PROBLEM ANALYSIS

Educational institutions receive a large number of student queries every day regarding course registration, class timetables, examination schedules, attendance records, library timings, hostel facilities, transportation services, and other campus-related information. These queries are usually handled manually by faculty members or administrative staff through phone calls, emails, or direct visits. This traditional approach is time-consuming and increases the workload of staff, especially during admission periods and examinations.

Students often experience delays in receiving information because administrative offices operate only during working hours. In addition, repetitive questions consume valuable time that could be used for other important academic and administrative tasks. As the number of students increases, managing these requests manually becomes more challenging and less efficient.

Artificial Intelligence provides an effective solution to this problem through conversational chatbots. An AI-powered chatbot can understand user queries written in natural language, identify the user's intent, and provide instant responses without human intervention. By using Natural Language Processing (NLP) and Natural Language Understanding (NLU)**,** the chatbot can accurately interpret different ways of asking the same question and respond appropriately.

In this project, the AI-Based Student Support Assistant is developed using Google Dialogflow to automate student support services. The chatbot is capable of answering frequently asked questions related to academic and campus services, ensuring that students receive quick, accurate, and consistent information at any time. This approach improves communication, reduces administrative workload, and enhances the overall student experience.

2.1 Existing System

The existing student support system mainly depends on manual communication between students and administrative staff. Students have to visit the office, make phone calls, or send emails to obtain information. This process consumes time and may result in delays, particularly during peak academic periods.

Disadvantages of the Existing System

- Time-consuming process
- Increased workload for staff

- Delayed responses
- Limited office hours
- Repetitive handling of similar queries
- Reduced efficiency during busy periods

2.2 Proposed System

The proposed system is an AI-Based Student Support Assistant developed using Google Dialogflow**.** The chatbot uses Artificial Intelligence techniques such **as** Natural Language Processing (NLP)**,** Intent Recognition**,** and Entity Extraction to understand student queries and provide appropriate responses.

The chatbot allows students to ask questions about course registration, timetables, examinations, attendance, and campus services in natural language. It responds instantly, improving communication and reducing the workload of administrative staff.

Advantages of the Proposed System

- 24×7 availability
- Instant responses to student queries
- Reduced administrative workload
- Improved communication
- Accurate and consistent information
- User-friendly interaction
- Easy to update and maintain

AGENT DESIGN

The AI-Based Student Support Assistant is designed using Google Dialogflow to provide instant responses to student queries. The chatbot uses Natural Language Processing (NLP) to understand user input, identify the correct intent, and generate an appropriate response. It is designed to improve communication between students and the institution by providing quick and accurate information related to academic and campus services.

Objective

The objective of the chatbot is to provide 24×7 assistance to students by answering queries related to course registration, timetable, examination schedule, attendance, and campus services.

Users

The primary users of the chatbot are:

- Students
- Faculty Members
- Administrative Staff

Environment

- Platform: Google Dialogflow ES
- Device: Laptop or Mobile
- Browser: Google Chrome
- Internet Connection

Inputs

The chatbot accepts natural language queries such as:

- How do I register for courses?
- Show my timetable.
- What is my attendance?
- When is my exam?

Outputs

The chatbot provides:

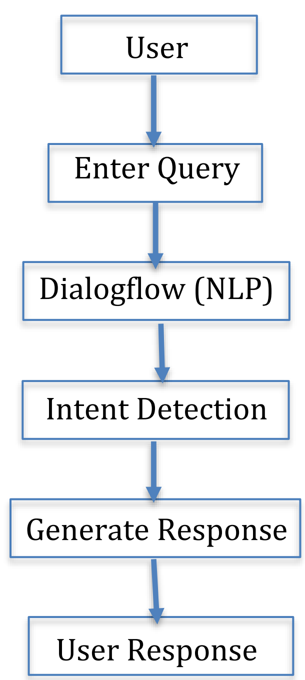
- Course registration details
- Timetable information
- Examination schedule
- Attendance details
- Campus service information

Performance Measures

The chatbot is evaluated based on:

- Response accuracy
- Fast response time
- Correct intent recognition
- User satisfaction
- Handling of unknown queries

Workflow



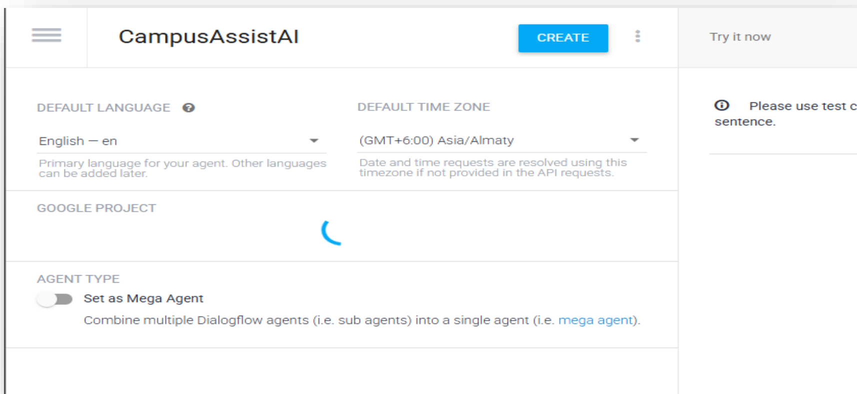
DIALOGFLOW AGENT DEVELOPMENT

Google Dialogflow ES was used to develop the CampusAssist AI chatbot. It is a cloud-based conversational AI platform that uses Natural Language Processing (NLP) to understand user queries and generate appropriate responses.

4.1 Agent Creation

A new agent named CampusAssist AI was created with English as the default language and Asia/Kolkata as the time zone.

📷 **Screenshot:**



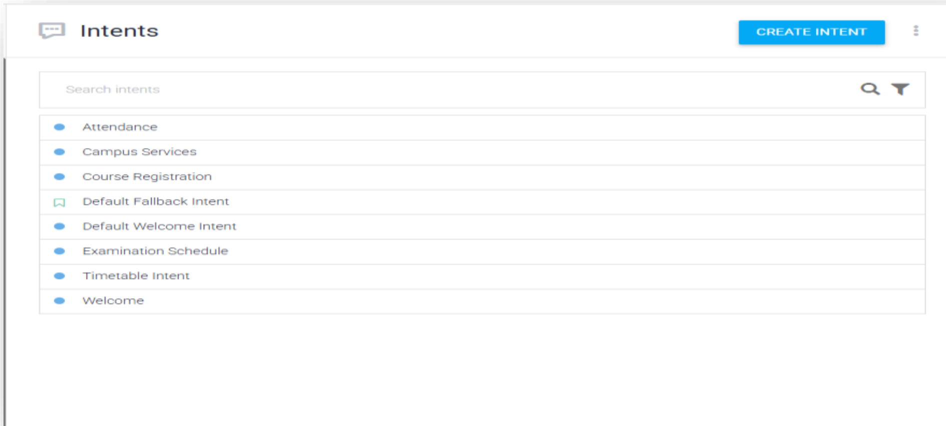
4.2 Intents

Intents were created to identify different student queries. The chatbot includes the following intents:

- Default Welcome Intent
- Course Registration
- Timetable
- Examination Schedule
- Attendance
- Campus Services
- Goodbye
- Default Fallback Intent

📷 **Screenshot:** Intents List

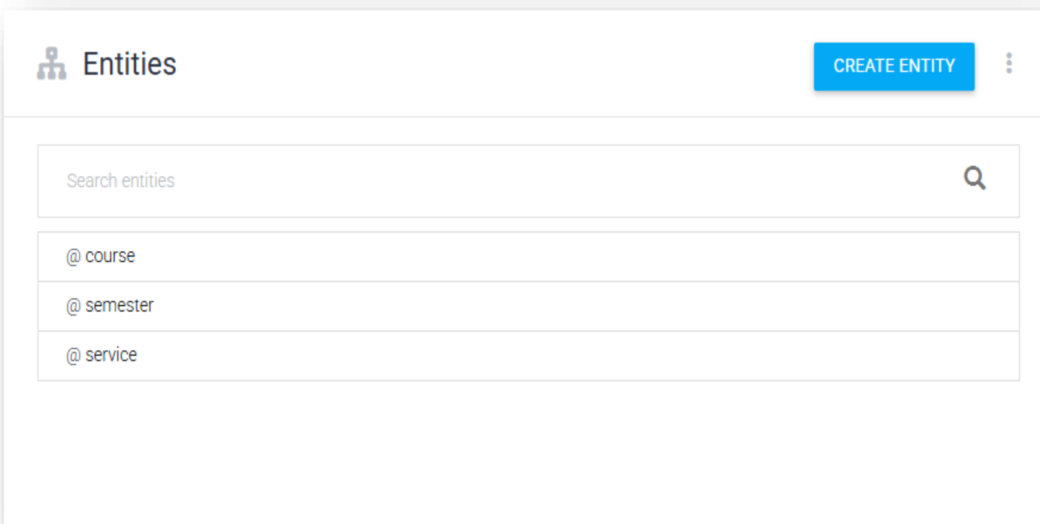
4.3



Entities

Entities were created to identify important information from user queries, such as Department and CampusService**.**

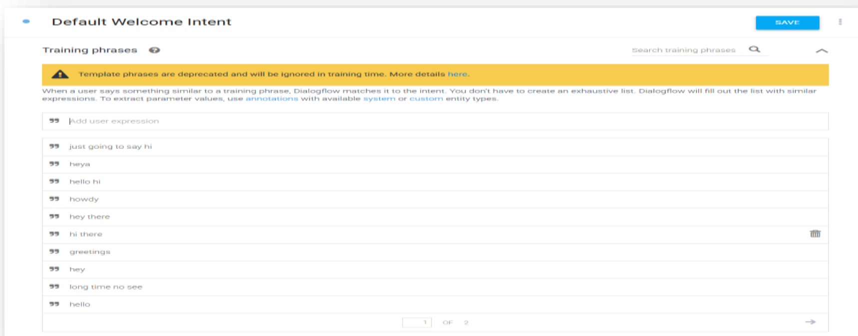
📷 **Screenshot:** Entities Page

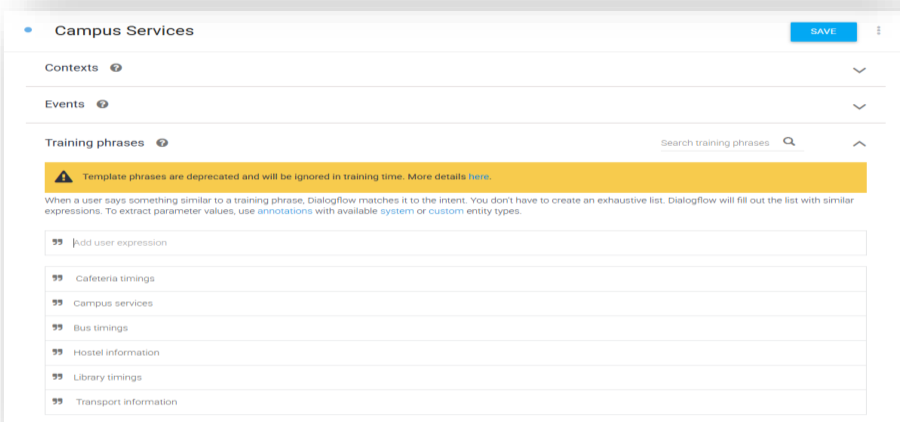


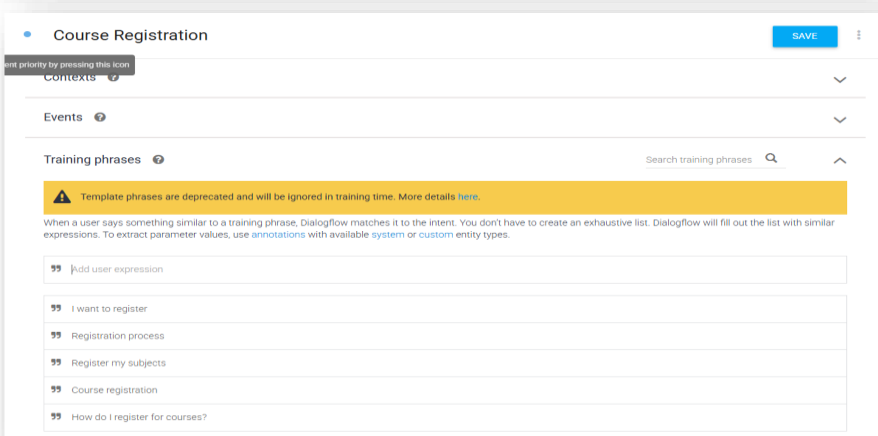
4.4 Training Phrases and Responses

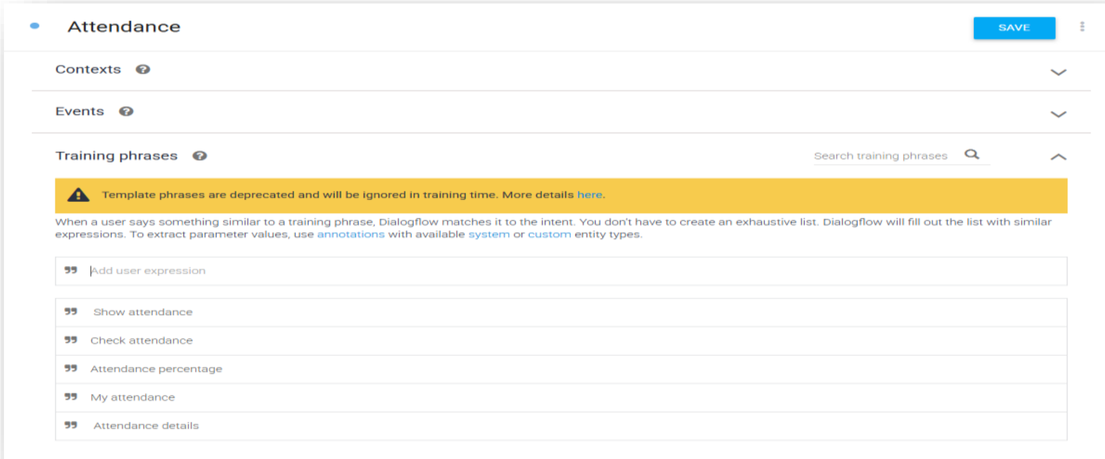
Each intent contains multiple training phrases to recognize different user queries. Appropriate responses were added to provide accurate information related to student services.

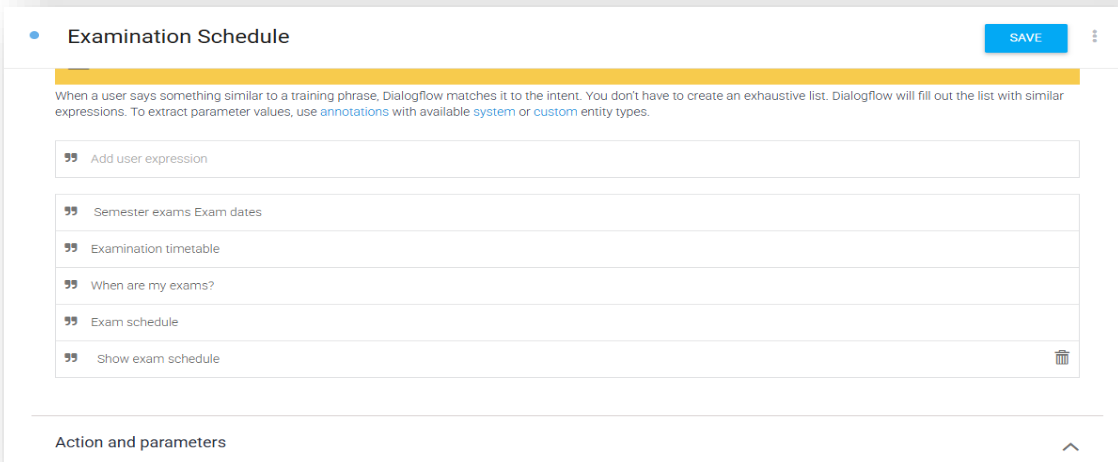
📷 **Screenshot:** Training Phrases and Responses

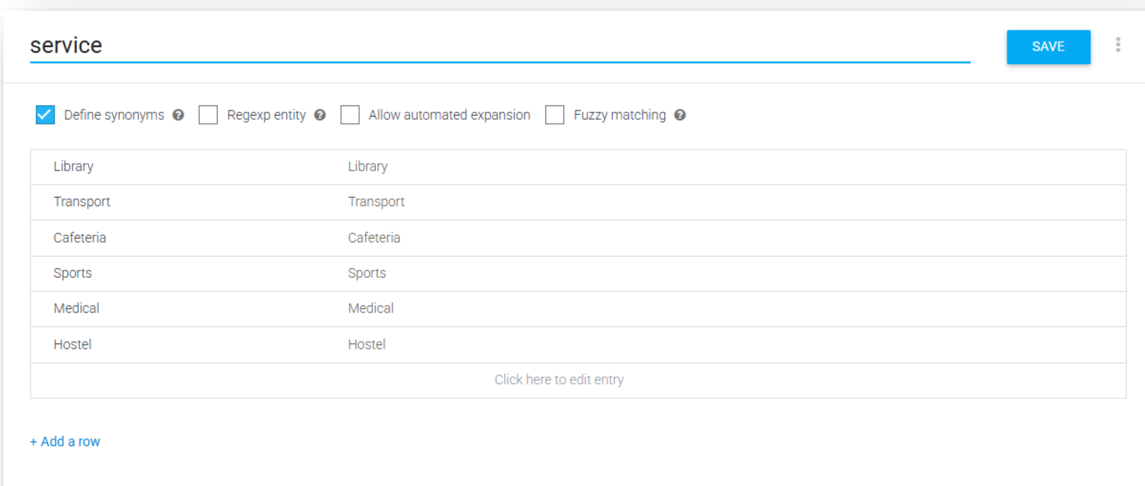


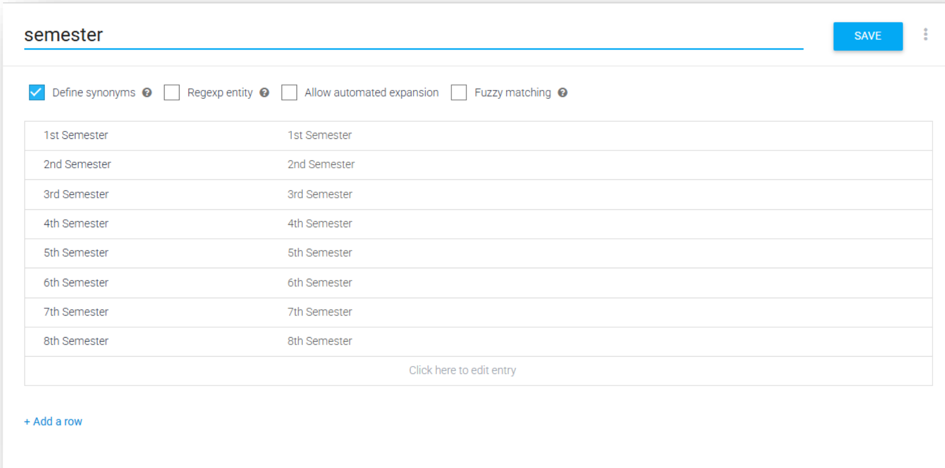


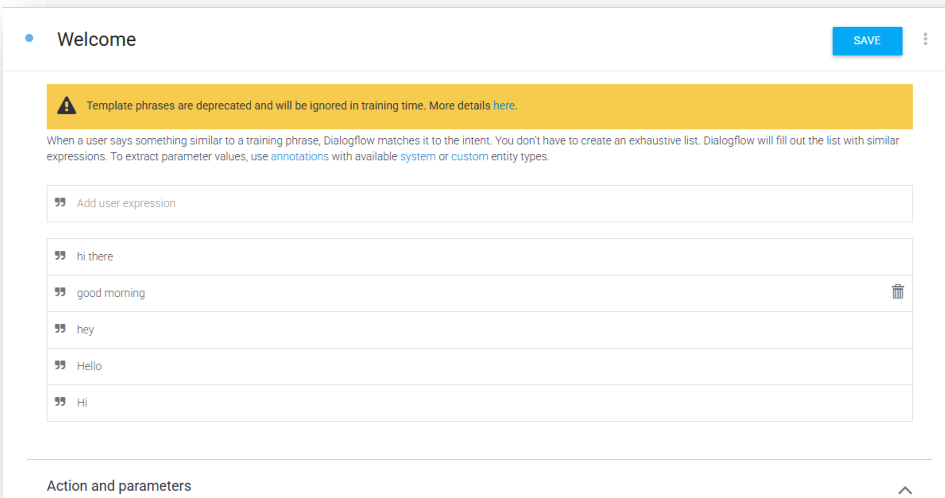



Attendance
SAVE
Contexts
Events
Training phrases
Search training phrases
Template phrases are deprecated and will be ignored in training time. More details here.
When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use annotations with available system or custom entity types.
Add user expression
Show attendance
Check attendance
Attendance percentage
My attendance
Attendance details


Examination Schedule
SAVE
When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use annotations with available system or custom entity types.
Add user expression
Semester exams Exam dates
Examination timetable
When are my exams?
Exam schedule
Show exam schedule
Action and parameters


service
SAVE
Define synonyms
Regexp entity
Allow automated expansion
Fuzzy matching
Library
Library
Transport
Transport
Cafeteria
Cafeteria
Sports
Sports
Medical
Medical
Hostel
Hostel
Click here to edit entry
+ Add a row


semester
SAVE
Define synonyms
Regexp entity
Allow automated expansion
Fuzzy matching
1st Semester 1st Semester
2nd Semester 2nd Semester
3rd Semester 3rd Semester
4th Semester 4th Semester
5th Semester 5th Semester
6th Semester 6th Semester
7th Semester 7th Semester
8th Semester 8th Semester
Click here to edit entry
+ Add a row


Welcome
SAVE
Template phrases are deprecated and will be ignored in training time. More details here.
When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use annotations with available system or custom entity types.
Add user expression
hi there
good morning
hey
Hello
Hi
Action and parameters

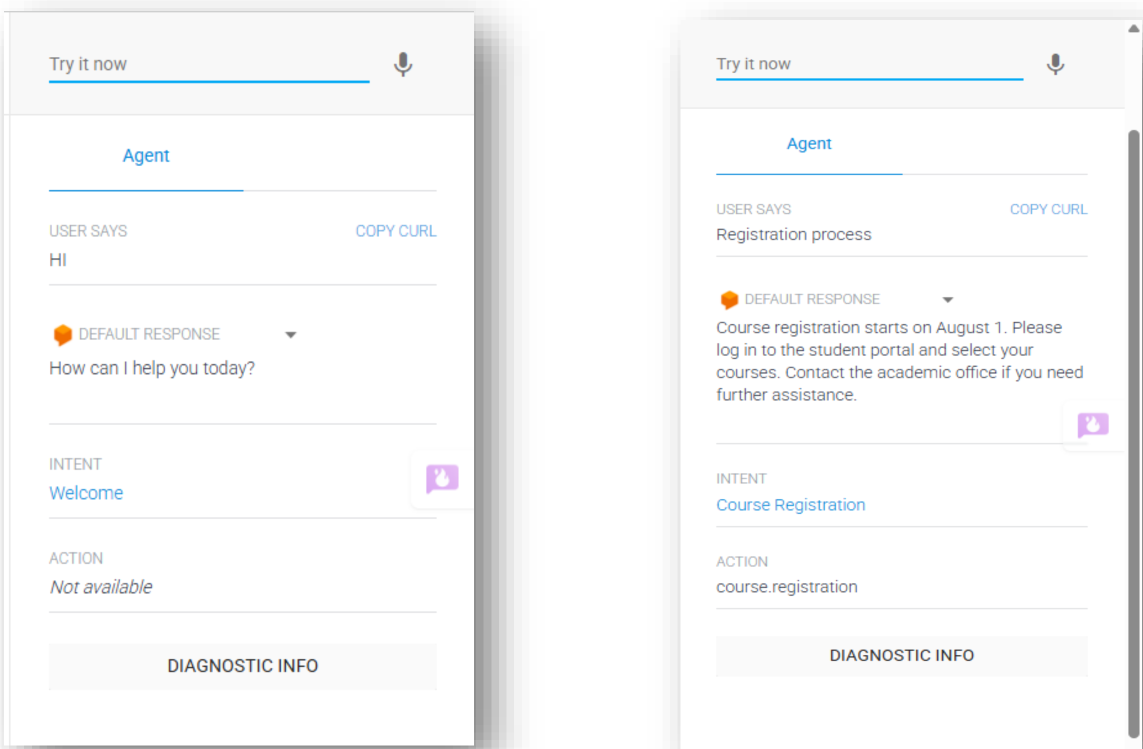
4.5 Responses

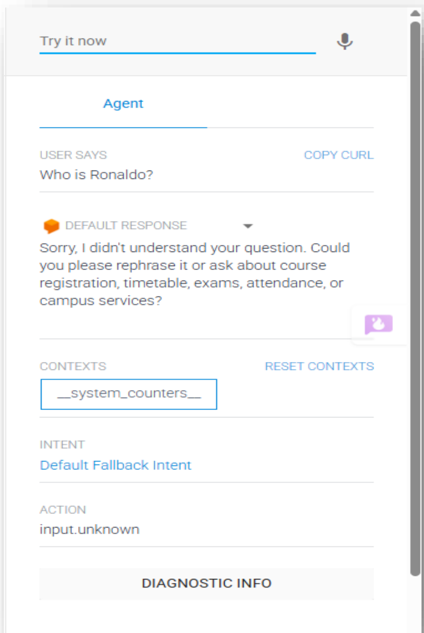
Each intent contains predefined responses that are displayed when the corresponding intent is detected.

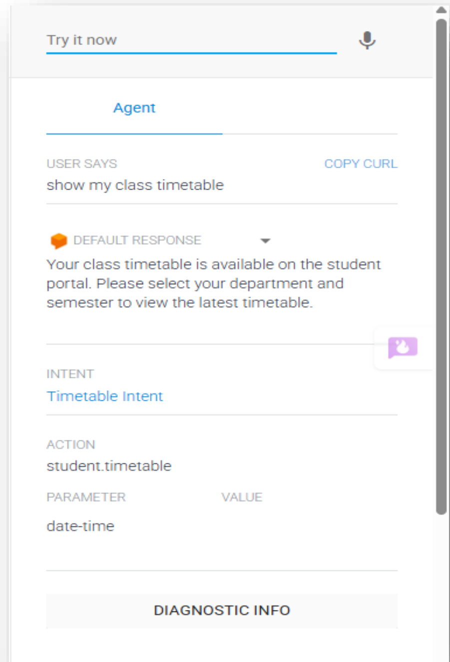
Example Responses:

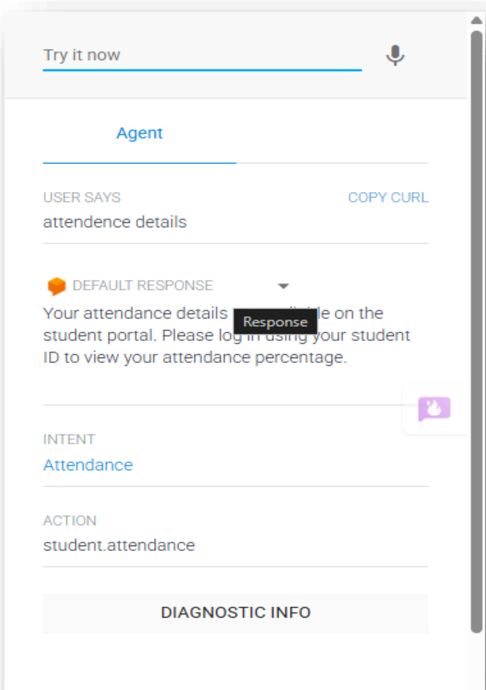
- Course registration begins on 5 August.
- Your attendance is 92%.
- Library timing is from 8:00 AM to 8:00 PM.
- Semester examinations begin on 20 September.

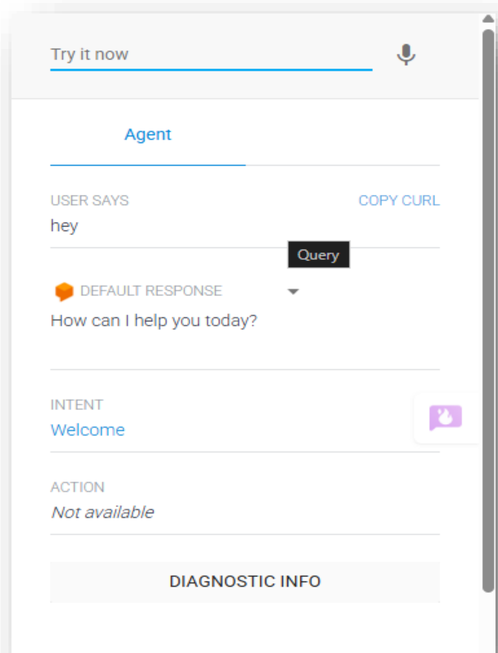
📷 **Screenshot 5:** Responses Section




Try it now
Agent
USER SAYS
COPY CURL
Who is Ronaldo?
DEFAULT RESPONSE
Sorry, I didn't understand your question. Could you please rephrase it or ask about course registration, timetable, exams, attendance, or campus services?
CONTEXTS
RESET CONTEXTS
__system_counters__
INTENT
Default Fallback Intent
ACTION
input.unknown
DIAGNOSTIC INFO


Try it now
Agent
USER SAYS
COPY CURL
show my class timetable
DEFAULT RESPONSE
Your class timetable is available on the student portal. Please select your department and semester to view the latest timetable.
INTENT
Timetable Intent
ACTION
student.timetable
PARAMETER
VALUE
date-time
DIAGNOSTIC INFO


Try it now
Agent
USER SAYS
COPY CURL
attendence details
DEFAULT RESPONSE
Your attendance details
Response
e on the
student portal. Please log
your student
ID to view your attendance percentage.
INTENT
Attendance
ACTION
student.attendance
DIAGNOSTIC INFO


Try it now
Agent
USER SAYS
COPY CURL
hey
Query
DEFAULT RESPONSE
How can I help you today?
INTENT
Welcome
ACTION
Not available
DIAGNOSTIC INFO

PROBLEM FORMULATION AND DECISION FLOW

Problem formulation is the process of defining how the chatbot solves a user's query. The chatbot receives a question from the user, identifies the appropriate intent, and generates a suitable response. The conversation continues until the user's query is resolved.

5.1 Initial State

The user opens the chatbot and enters a query related to student services.

5.2 Goal State

The chatbot provides the correct information requested by the user and successfully completes the conversation.

5.3 Possible Actions

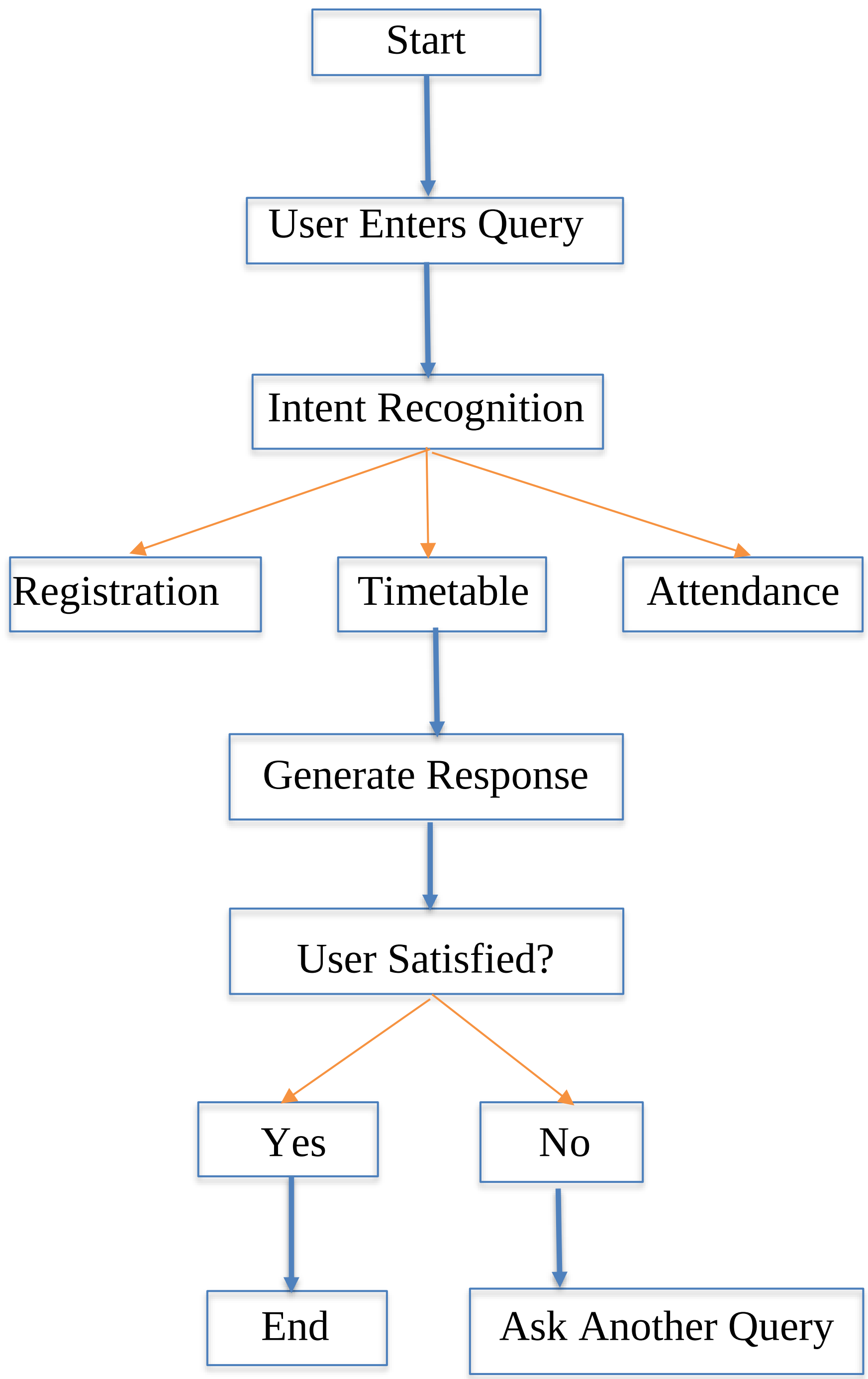
The chatbot performs the following actions:

- Receives the user's query.
- Identifies the intent.
- Extracts relevant information (if required).
- Generates an appropriate response.
- Continues the conversation or ends it.

5.4 State Transition

The chatbot moves through different stages from receiving the query to generating the final response.

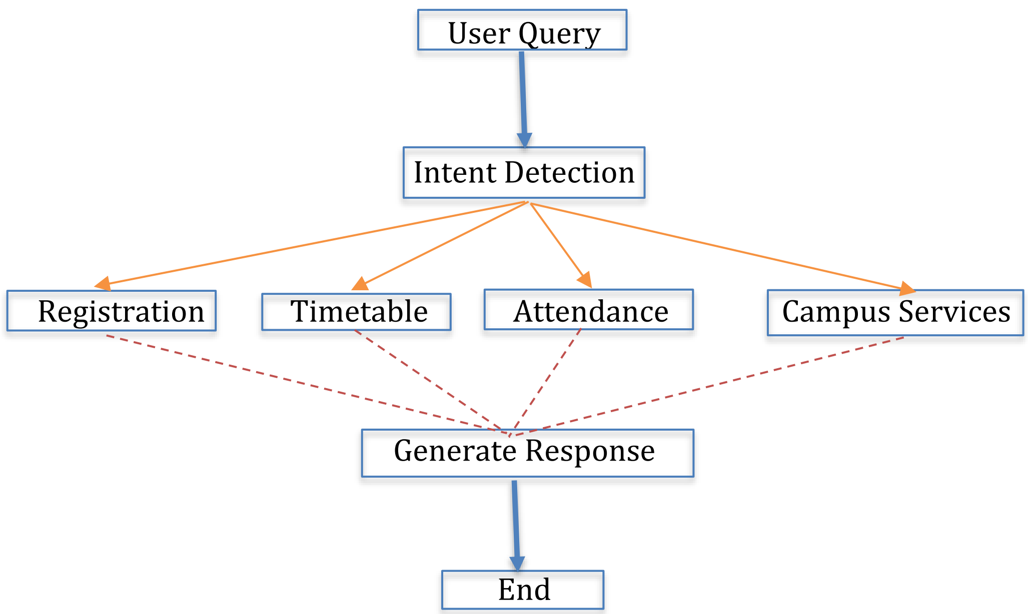
Conversation Flow Diagram:



SEARCH STRATEGY APPLICATION:

Search strategies in Artificial Intelligence help a system find the correct solution from multiple possible options. In this project, the chatbot uses Intent Recognition to identify the user's query and provide the most appropriate response.

For this chatbot, the Breadth First Search (BFS) approach can be used to represent the conversation flow. BFS explores all possible intents at the current level before moving to the next level. This helps the chatbot quickly identify the correct intent based on the user's input.



TESTING AND EVALUATION

The developed CampusAssist AI chatbot was tested using different student queries to verify its functionality and response accuracy. The chatbot successfully recognized the user's intent and generated appropriate responses for most of the queries. The testing confirmed that the chatbot provides quick and relevant information related to student services.

Test Cases

Test Case	User Query	Expected Result	Status
1	Hi	Welcome message	Pass
2	How do I register for courses?	Registration details	Pass
3	Show my timetable	Display timetable	Pass
4	When is my exam?	Examination schedule	Pass
5	What is my attendance?	Attendance details	Pass
6	Library timing	Library information	Pass
7	Bye	Goodbye message	Pass

Evaluation

The chatbot was evaluated based on the following parameters:

- **Accuracy:** Correctly identified user intents for most of the test queries.
- **Response Time:** Generated responses within a few seconds.
- **Response Quality:** Responses were clear, relevant, and easy to understand.
- **Unknown Queries:** The Default Fallback Intent handled unsupported questions by asking the user to rephrase them.

CONCLUSION

The AI-Based Student Support Assistant developed using Google Dialogflow successfully demonstrates how Artificial Intelligence can improve student support services in educational institutions. The chatbot effectively understands user queries using Natural Language Processing (NLP) and provides quick, accurate, and relevant responses related to course registration, class timetables, examination schedules, attendance, and campus services.

The implementation of intents, entities, training phrases, and contexts enables the chatbot to interact naturally with users and handle different types of student queries efficiently. The testing results show that the chatbot performs well in recognizing user intents and generating appropriate responses, thereby reducing the workload of administrative staff and improving communication with students.

Overall, the project highlights the practical application of conversational AI in education by providing a 24×7, user-friendly, and efficient support system**.** In the future, the chatbot can be enhanced by integrating it with the college database, adding multilingual and voice support, and providing personalized student services, making it a more intelligent and comprehensive virtual assistant for educational institutions.

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